

Question ID: 69f609b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

How many solutions does the equation $12(x - 3) = -3(x + 12)$ have?

Answer

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Correct Answer: A

Rationale

Choice A is correct. Distributing **12** on the left-hand side and **—3** on the right-hand side of the given equation yields $12x - 36 = -3x - 36$. Adding **3x** to each side of this equation yields $15x - 36 = -36$. Adding **36** to each side of this equation yields $15x = 0$. Dividing each side of this equation by **15** yields $x = 0$. This means that **0** is the only solution to the given equation. Therefore, the given equation has exactly one solution.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.